ResQ - Disaster Alert & Resource Coordination System
Product Requirements Document (PRD)

1. Overview

ResQ is a serverless disaster-response coordination platform designed for flood, landslide, fire, and emergency situations. The system focuses on coordination between district authorities, responders, relief camps, and volunteers. Citizens can report incidents through web APIs or SMS fallback.

2. Goals

Real-time incident reporting, resource tracking, automated escalation of unresolved incidents, district-level coordination, evidence uploads, and emergency notifications.

3. Users

Citizen, Responder, District Admin, Relief Camp Coordinator.

4. Functional Requirements

Incident reporting, resource registration, assignment workflows, resolution workflows, authentication, notifications, escalation engine, evidence uploads.

5. Data Model

Incidents: districtID, incidentID, timestamp, reporterID, severity, location, status. Resources: districtID, resourceID, type, capacity, availability. Users: userID, role, district, contact details.

6. API Endpoints

POST /incident/report GET /incident/{district} PATCH /incident/{id}/assign PATCH /incident/{id}/resolve POST /resource/register GET /resource/{district} PATCH /resource/{id}/status POST /auth/register POST /auth/login

7. Event-Driven Features

Auto-escalation after 30 minutes, severity-5 broadcasts, SMS ingestion pipeline, scheduled monitoring jobs.

8. Free-Tier Friendly Tech Stack

Go, AWS Lambda, API Gateway, DynamoDB Free Tier, SNS, EventBridge, S3, GitHub Actions, Postman.

9. Non-Functional Requirements

High availability, stateless architecture, secure JWT authentication, scalable serverless infrastructure.

10. Resume Description

Built a fully serverless disaster-response backend supporting incident management, resource coordination, automated escalation, and emergency notifications.

UI / Design Strategy

Instead of building custom UI from scratch, use community Figma templates for emergency dashboards, admin panels, logistics management systems, or disaster monitoring dashboards. Adapt existing designs and focus engineering effort on backend architecture.